Lab II:
Run a simple C application

Intel x86 Bootstrap

Objective: Configure the vCPU to support x86 64bits user application

- Memory
  - Segmentation (global descriptor table or GDT) & Paging (page table)
  - Permissions (read/write)
- System Calls (syscalls)
  - Syscalls routine in kernel mode
  - `syscall` instruction
- Mode utilisateur
  - Segments update
  - Registers configuration
  - `sysret` instruction

Compilation et chargement

Objectif: Compile and Load Guest Code (boot + application) in Memory

- Compilation
  - Dynamic Linking (by default): application code only
  - Static Linking: application code + library
  - Library: Musl (lighter than GlibC)
- Loading
  - Link Edition: associate an address to a symbol (fonction, variable, ...)
  - Copying the binary segments in guest physical memory

Gestion des appels système

Objective: Handling guest application syscalls

- “Empty” kernel
  - No syscall treatment
  - VMM catches the guest syscall and forwards them (when necessary) to the host
- VMM Handling
  - Identifying the arguments (Application Binary Interface)
  - Perform the corresponding operation